ISA5810 DM2026 Lab 1 — Master Notebook Questions

Fill in this file as you work through DM2026-Lab1-Master.ipynb. Submit this completed file converted to PDF together with the rest of your Lab 1 work.

Please do not change the style of this document.

Question numbers 1–10 follow the same order as the notebook (top to bottom).

Student Information

Please complete the fields below. For **English Name** and **Chinese Name**, you may fill in **only one** if you do not have the other.

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Instructions

These questions go with the Master Jupyter notebook (DM2026-Lab1-Master.ipynb). Each question is tied to a specific notebook section. Open the notebook, find **Question N (take home)**, and write your answer here.

We **strongly** suggest running the notebook to learn the lab, but the TAs will **not** audit whether you executed every cell to answer the questions in this document.

How to answer each question

1. Find the **notebook section** listed under the question (use the Table of Contents if needed).
2. Read the task; change parameters when asked (e.g. stop_words, thresholds, k, minSup).
3. Write a clear answer in the **Answer** space below each question.
4. Leave the question text unchanged; write only in the answer and disclosure fields.

Sources, honesty, and AI use

Acceptable vs. Unacceptable AI Usage: Generative AI should serve as an interactive tutor to guide your understanding, not as an automated solver completing the assignment for you.

- **Permitted (AI as Guide & Editor):**
  - Asking AI to explain unfamiliar theoretical concepts, discuss algorithmic trade-offs, or troubleshoot runtime errors in the notebook.
  - Polishing the grammar, phrasing, and readability of your own original, self-written drafts.

- **Prohibited (AI as Solver):**
  - Copying assignment questions directly into an AI tool to generate answers.
  - Using AI to fabricate experimental results or observations without executing the code in DM2026-Lab1-Master.ipynb.
  - Submitting machine-generated prose wholesale.

- **External material:** If you use websites, books, papers, or documentation beyond the notebook, cite them with **clickable links** (or DOI/ISBN when no URL exists).

- **When AI was used:** Your answer is acceptable if you (1) **disclose** AI use and (2) include **at least one external link** that validates the information in your answer (official docs, textbook, paper, or reputable tutorial — not the AI chat alone as the only source).

- **When AI was not used:** You may write "No" under **AI used** or leave it blank only if you truly did not use AI for that question. References are still encouraged if you looked anything up.

Under each question, complete:

- **AI used for this answer:** Yes / No
- **If yes, how:** (e.g. brainstorming, grammar, concept explanation)
- **External references (links):** *(required when AI = Yes; encouraged otherwise)*

Write in English. Keep answers concise and specific.

Questions

Question 1: Data Filtering & Domain Relevance

Notebook: Section 3.2 — Question 1 (take home)

Task: Filter the dataframe **X** for the category **sci.med** and check its length.

Questions:

1. How many documents are in this specific category?
2. From a data mining perspective, why is isolating specific categories a crucial first step before applying algorithms like frequent pattern mining?

Answer:

1. 594 documents.

2. Let's say you want to find words that appear together in a medical article, such as "doctor," "patient," and "treatment." If you mix in words related to cloth, you'll end up with a whole bunch of combinations that have nothing to do with medicine.

AI used for this answer: NO

If yes, how:

External references (links):

Question 2: Handling Missing Data

Notebook: Section 4.1 — Question 2 (take home) (after the **dropna()** demonstration)

Task: In Section 4.1, a dummy record with a missing value is injected and then removed using the **dropna()** function.

Question: If 40% of documents were missing the **category_name** label, why would simply dropping these rows (with **dropna()**) be a flawed data mining strategy? Name one objective alternative so the text data can still be utilized.

Answer:

1 . Using dropna() to directly delete data with missing labels may result in the deletion of useful data, thus altering the distribution of the original data.

2. Unsupervised Learning

AI used for this answer: No

If yes, how:

External references (links):

Question 3: Data Sampling Risks

Notebook: Section 5.1 — Question 3 (take home)

Task: In Section 5.1, the code uses **X_sample = X.sample(n=1000)** to create a subset. Compare the bar chart of the original category distribution to the sampled distribution.

Questions:

1. Do the exact category proportions perfectly match the original?
2. What is the specific risk of simple random sampling on a highly imbalanced dataset?

Answer:

1. **No, not exactly.**

2. Underrepresentation、Sampling Bias / Sampling Variability 。If the dataset is imbalanced, random sampling may include too few documents from the minority categories. This can make the sample less representative of the original dataset.

AI used for this answer: No

If yes, how:

External references (links):

Question 4: Text Vectorization and Matrix Sparsity

Notebook: Section 5.3 — Question 4 (take home)

Task: In Section 5.3, the **CountVectorizer** converts text into a document-term matrix using **stop_words='english'**. Change this parameter to **stop_words=None** and re-run the cell to check the shape of **X_counts**.

Questions:

1. Does the total number of features (columns) increase or decrease?
2. Since stopwords are highly common words that appear in almost all documents, what effect does adding them back have on the overall sparsity (the percentage of zeros) of the matrix?

Answer:

1. **Increases the number of features**

2. Common stopwords appear in many documents、more non-zero entries、lower percentage of zeros

AI used for this answer: No

If yes, how:

External references (links):

Question 5: Feature Selection — Variance Thresholding

Notebook: Section 5.3.1 — Question 5 (take home)

Task: In the variance filtering section, re-run **variance_filter** using threshold values **threshold = 0.0, threshold = 0.01, and threshold = 0.05**. Record the number of features marked as **keep == True**.

Questions:

1. What happens to the total size of the vocabulary as the variance threshold increases from 0.0 to 0.05?
2. In terms of document frequency, what kinds of words have variance close to zero and are eliminated first?
3. Why is eliminating near-zero variance features considered a safe way to reduce dimensionality without harming model accuracy?

Answer:

1. The vocabulary size decreases significantly

2. Rare words and nearly constant words

3. It reduces dimensionality and computational cost.

Threshold	Number of terms kept (keep == True)
0.0	35482
0.01	5089
0.05	1491

AI used for this answer: No

If yes, how:

External references (links):

Question 6: Feature Correlation — Linear (Pearson) vs. Monotonic (Spearman)

Notebook: Section 5.3.2 — Question 6 (take home)

Task: Run the correlation filter against the binary indicator for the **comp.graphics** class. Inspect the top 10 features sorted by **pearson_r** and compare them to the top 10 features sorted by **spearman_r**.

Questions:

1. Are the top-correlated words generic stop words or domain-specific terminology for graphics?
2. What is the mathematical difference in what **Pearson** correlation measures versus what **Spearman** rank correlation measures?
3. In what specific scenario would a word have a high **Spearman** rank correlation with a topic, but a significantly lower **Pearson** correlation?

Answer:

1. domain-specific terminology for graphics

2. Pearson correlation measures the linear relationship between two variables using their original values. Spearman correlation uses ranks to measure whether two variables have a monotonic relationship. Compared with Pearson, Spearman is less affected by extreme values.

3. Pearson correlation can be skewed by outliers if a few documents have extremely high counts of a graphics-related word. Spearman remains robust by using ranks instead of raw counts.

AI used for this answer: No

If yes, how:

External references (links):

Question 7: Frequent Pattern Mining — Algorithm & Parameter Dynamics

Notebook: Section 5.4.2 — Question 7 (take home)

Task: In the Frequent Pattern Mining section, run **Top-K FAE** with **k = 100 and k = 500**. Then, run **MaxFPGrowth** using minimum support thresholds **minSup=9 and minSup= 3**.

Questions:

1. How does decreasing **minSup** (from 9 to 3) in **MaxFPGrowth** affect the execution runtime and the total number of discovered patterns?
2. How does the parameter k in **Top-K FAE** differ fundamentally from **minSup** in **MaxFPGrowth** when deciding which patterns to keep?
3. Why are "maximal" frequent patterns structurally more concise than standard frequent patterns?

Answer (1) — Runtime and pattern count when lowering minSup:

When minSup decreased from 9 to 3, the number of discovered maximal frequent patterns increased from 710 to 9,547, while the execution runtime increased from 0.0856 to 1.7702 seconds. This happened because a lower minimum support threshold allows more low-frequency itemsets to qualify as frequent, expanding the search space and increasing computational cost.

Answer (2) — Top-K k vs MaxFPGrowth minSup:

The parameter k in Top-K FAE specifies the number of highest-ranked frequent patterns to discover. In contrast, minSup in MaxFPGrowth specifies the minimum occurrence frequency required for a pattern to be considered frequent. Therefore, k controls the number of selected patterns, while minSup controls the frequency threshold. MaxFPGrowth additionally retains only maximal frequent patterns.

Answer (3) — Maximal vs standard frequent patterns:

Maximal frequent patterns are more concise because they only keep frequent itemsets that are not subsets of any larger frequent itemset. For example, if is a maximal frequent pattern, its subsets such as , , and do not need to be listed separately. This reduces redundancy and the total number of reported patterns. However, the exact support values of the omitted subsets are not preserved.

AI used for this answer: Yes

If yes, how: I used ChatGPT to help debug Python errors, understand the differences between Top-K FAE and MaxFPGrowth, interpret the experimental results, and improve the clarity and grammar of my English explanations. I ran the experiments myself in Jupyter Notebook and collected the results.

External references (links): PAMI — Top-K FAE

Question 8: Dimensionality Reduction — Motivation

Notebook: Section 5.5 — Question 8 (take home) (before PCA/t-SNE code)

Task: After text vectorization, the **X_counts** matrix creates tens of thousands of distinct features (words).

Questions:

1. If we want to visually map how these documents group by category on a scatter plot, why is it mathematically and visually impossible to plot the raw **X_counts** matrix directly?
2. What specific category of data mining algorithms (referenced in the helper files) must be applied first?

Answer:

1. **X_counts has too many features to display directly in a 2D or 3D scatter plot.**
2. **The required algorithms are dimensionality reduction algorithms, such as PCA (Principal Component Analysis), t-SNE (t-distributed Stochastic Neighbor Embedding), and UMAP (Uniform Manifold Approximation and**

Projection). These methods help visualize document similarities and potential category clusters in a lower-dimensional space.

AI used for this answer: Yes

If yes, how: I used ChatGPT to understand the limitations of visualizing high-dimensional text data, learn about dimensionality reduction algorithms, and improve my English explanation.

Externalreferences (links): https://scikit-learn.org/stable/modules/decomposition.html

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Question 9: Dimensionality Reduction — Linear (PCA) vs. Non-linear Manifolds (t-SNE/UMAP)

Notebook: Section 5.5 — Question 9 (take home) (after 2D/3D plots)

Task: In the 2D/3D Dimensionality Reduction section, compare the scatter plot projection generated by **PCA** against that generated by **t-SNE** or **UMAP**.

Questions:

1. Which method **(PCA or t-SNE/UMAP**) shows distinct, separated cluster islands for the 4 document categories, and which shows heavy overlap around the center?
2. Why does **PCA** struggle to separate high-dimensional text categories compared to **t-SNE** or **UMAP**?
3. Why must data analysts be careful not to over-interpret the physical distance between separate clusters in a **t-SNE** plot?

Answer:

1. **UMAP shows clearer separation between some of the four document categories. PCA has heavy overlap near the center, while t-SNE still shows many mixed points. However, none of the methods completely separates all four categories.**
2. **PCA is a linear method that tries to keep the most variance in the data. However, high-dimensional text data has complex relationships that PCA cannot easily capture. t-SNE and UMAP can preserve local relationships better, so they may show clearer clusters.**
3. **t-SNE mainly focuses on keeping similar points close together. The distance between different clusters may not represent their real distance in the original data. Therefore, we should not assume that two clusters are very different just because they look far apart.**

AI used for this answer: Yes

If yes, how: I used ChatGPT to help explain the differences between PCA, t-SNE, and UMAP and to organize my answers in English.

External references (links): https://scikit-learn.org/stable/modules/manifold.html

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Question 10: One-Hot Binarization

Notebook: Section 5.6 — Question 10 (take home)

Task: In Section 3.1, examine **X['category']** (integers 0, 1, 2, 3). Then look at the output of **LabelBinarizer** in Section 5.6, where each category is converted into a binary vector (e.g., [1, 0, 0, 0]).

Questions:

1. How many binary columns are created to represent the 4 categories?
2. If an algorithm (like linear regression or KNN) is given the raw integer column **(0, 1, 2, 3)** as input, what false mathematical assumption does it make about the categories?
3. How does one-hot binarization eliminate this false assumption?

Answer:

1. **Four binary columns are created to represent the four categories. Each category has one column with a value of 1, while the other columns have a value of 0.**

Original catagory	One-Hot
0	**[1, 0, 0, 0]**
1	**[0, 1, 0, 0]**
2	**[0, 0, 1, 0]**
3	**[0, 0, 0, 1]**

2. **Using raw integers (0, 1, 2, 3) may cause algorithms to assume that the categories have a meaningful order and numerical distance. For example, category 3 may appear larger than category 1, even though the numbers are only labels.**
3. **One-hot binarization converts each category into a separate binary vector. This removes the false numerical order and makes all different categories equally distant in the encoded space.**

AI used for this answer: NO

If yes, how:

External references (links):